Sanjith Shanmugathashan

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DSP/FPGA engineer working on PHY-layer development for communication systems, from signal-processing models to working hardware. I'm excited by challenging signal-processing problems and the insight gained from testing ideas in real systems.

Experience

- **Thakshana Technologies** — Colombo, Sri Lanka
  Team Lead DSP / FPGA (R&D) — *Jan, 2024 – Present*
  - Lead and mentor 9 engineers across DSP algorithm development and FPGA implementation, guiding architecture, reviewing designs and code, and setting engineering practices.

  Secure Satellite Modem (DVB-RCS2, Mesh Topology)
  - Owned the development of spread-spectrum waveforms: defined modulator and demodulator architectures and coordinated implementation across the team. Brought the burst detector up on DDR-backed deep buffering to hold spread bursts. Drove acquisition and synchronization down to -11 **dB SNR**. Resolved failure modes that surfaced only at full receiver-chain integration.
  - Implemented the **jammer detector** for protected waveforms, operating at up to **30 dB JSR** and built for **120 MSa/s**. Laid the groundwork it runs on: hop-synchronized RMS information carried through the burst detector, per-hop jammer score reporting, and waveform-dependent thresholds controlled by higher layers. Tested it in the field against a jammer and resolved the issues that surfaced.
  - Took over design and maintenance of the modulator and demodulator for the modem, covering **32+ waveforms** and frequency-hopped streams. Designed its core blocks—carrier frequency and phase recovery, and turbo FEC decoding—and distributed the remaining blocks across the team. Integrated the chain and validated it over digital loopback. Built a regression setup to support verification as the design evolves. Met **80 MSym/s** demodulator rate, **100 Mb/s** output rate, and $<$**50** μs modulator latency requirements, with end-to-end backpressure support.
  - Led development of the PS–PL data-movement subsystem: a customized AXI-DMA and a PetaLinux driver on hugepage-backed buffers. It carries variable-length packet streams and sustains **758 Mb/s** even at the worst-case 64-byte packet size.
  - Built two stages of a receive front-end resampling chain in VHDL: a Xilinx FIR IP filter and a hand-written double-rate half-band decimator. Integrated and verified the full chain — fractional resampler and CIC included — resampling a fixed **491.52 MSa/s ADC** stream to any rate from **0.5 to 120 MSa/s**, with timing and AGC gain metadata carried alongside.
  - Designed the Digital Automatic Gain Control for the **DVB-S2** forward-link receiver of the satellite modem: a double feedback-loop structure sustaining **120 MSa/s** that holds gain stable against jammers up to **30 dB JSR**, and characterized convergence, threshold and stability across symbol rates alongside the algorithm team.

  Waveform Development for SDR
  - Designed an end-to-end MATLAB model of **Zak-OTFS** with a linear-complexity equalizer, demonstrating improved BER performance over CP-OFDM under high-Doppler TDL channel conditions. Validated the $\sim$1 MHz, RRC-shaped waveform on an Ettus USRP through RF loopback and over-the-air experiments. Now investigating link budget, power-amplifier nonlinearity and in-band distortion, with channel-emulator validation.
  - Developed a feedback **Kalman equalizer** in MATLAB for **MIL-STD-188-110B** over Watterson-fading HF channels, benchmarked against the figures in the standard inside the integrated channel model. Implemented it in C++ on an Arm Cortex-A53 with headroom to **40 kSym/s** for planned 110C wideband operation, validated that implementation against MATLAB vectors, and supported integration.
  - Owned RTL design of the **GMSK** and **Narrowband HF Hop** modulators: derived the fixed-point MATLAB model from the floating-point reference to settle quantization, set the top-level plan and core structure the team built to, then handled integration and supported delivery debug.
  - Led exploratory and bring-up work beyond the main datapath: **DFX** experiments swapping FPGA modules at runtime with millisecond-scale reconfiguration, performance evaluation of a DDR-backed interleaver (reaching $\sim$**80 Mb/s**), and bring-up of the AFE7444 RF front end over **JESD204B**.

- **Wadduwage Lab** — Old Dominion University, VA
  Undergraduate Researcher – Computational Imaging — *Aug, 2023 – Jun, 2024*
  - Led the team in simulating a novel 3D computational fluorescence microscope that encodes volumetric information onto a 2D detector using structured illumination generated by a Diffractive Deep Neural Network.
  - Studied ill-posed estimation problems. Analyzed sensitivity to noise, model mismatch and stability–fidelity trade-offs.
  - Utilized the Plug-and-Play reconstruction framework for 3D volumes using 2D denoisers applied along multiple planes (e.g. xy, yz, zx), reconstructing a $32 \times 32 \times 8\,\mu$m volume at $8\times$ temporal compression.
- **Singapore University of Technology and Design** — Singapore
  Research Assistant Fellow – BLE AoA Localization — *Jan, 2023 – Aug, 2023*
  - Prototyped a Bluetooth Low Energy (BLE) angle-of-arrival (AoA) based localization system, reaching **centimeter-level** positioning accuracy from direction estimation and spatial inference.
  - Implemented and evaluated signal processing pipelines for AoA estimation, including phase-based measurements across antenna elements and sensitivity analysis to noise, multipath and hardware impairments.

Skills

- Programming : ***VHDL***, ***MATLAB***, *Embedded C/C++*
- Tools : ***Xilinx Vivado***, *Questa Simulator*, ***PetaLinux***, *Altium, Git*

Education

- **University of Moratuwa** — Moratuwa, Sri Lanka
  BSc (Electronic and Telecommunication Engineering) — *Jan, 2020 – Jun, 2024*
  CGPA : 4.12/4.2; Gold Medalist
- **J/Hartley College** — Jaffna, Sri Lanka
  GCE Advanced Level — *Jan, 2011 – Aug, 2018*
  3A's; ***6th out of about 100,000****; Z-score 2.7266*

Awards/ Achievements

- **Robofest - SINDiB** — *Sep, 2023*
  Sri Lanka Institute of Information Technology (SLIIT) - 2nd runners-up
- **PCB Design & Fabrication** — *Dec, 2020*
  IEEE Student Branches of University of Peradeniya & Open University of Sri Lanka - Won

Certifications

- Zak-OTFS and Delay-Doppler Processing for Next-Gen Communications — *IEEE: Oct, 2025*
- Designing with AXI DMA and using AXI DMA under Linux — *Green-Electrons: Apr, 2025*
- Designing FPGAs Using Vivado Design Suite 1 — *AMD: Jul, 2024*
- Designing with VHDL — *AMD: Jul, 2024*